



Bikkad IT Institute

05. SQL JOINS

Compiled by Bikkad IT Institute | Teaching Notes

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SQL JOINS

05 · Joins — Complete Reference Guide

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1. Basic Joins

→ *No Join — retrieve customers and orders as separate results*

```
SELECT * FROM customers;  
SELECT * FROM orders;
```

→ *INNER JOIN — only customers who have placed an order*

```
SELECT  
    c.id,  
    c.first_name,  
    o.order_id,  
    o.sales  
FROM customers AS c  
INNER JOIN orders AS o  
    ON c.id = o.customer_id
```

→ *LEFT JOIN — all customers, including those without orders*

```
SELECT  
    c.id,  
    c.first_name,  
    o.order_id,  
    o.sales  
FROM customers AS c  
LEFT JOIN orders AS o  
    ON c.id = o.customer_id
```

→ *RIGHT JOIN — all orders, including those without matching customers*

```
SELECT
```

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```
c.id,  
c.first_name,  
o.order_id,  
o.customer_id,  
o.sales  
FROM customers AS c  
RIGHT JOIN orders AS o  
ON c.id = o.customer_id
```

→ *Alternative to RIGHT JOIN — rewrite using LEFT JOIN*

```
SELECT  
c.id,  
c.first_name,  
o.order_id,  
o.sales  
FROM orders AS o  
LEFT JOIN customers AS c  
ON c.id = o.customer_id
```

→ *FULL JOIN — all customers and all orders, even if no match*

```
SELECT  
c.id,  
c.first_name,  
o.order_id,  
o.customer_id,  
o.sales  
FROM customers AS c  
FULL JOIN orders AS o  
ON c.id = o.customer_id
```

2. Advanced Joins

→ *LEFT ANTI JOIN — customers who have NOT placed any order*

```
SELECT *  
FROM customers AS c  
LEFT JOIN orders AS o  
ON c.id = o.customer_id  
WHERE o.customer_id IS NULL
```

→ *RIGHT ANTI JOIN — orders with no matching customers*

```
SELECT *  
FROM customers AS c  
RIGHT JOIN orders AS o  
ON c.id = o.customer_id  
WHERE c.id IS NULL
```

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→ *Alternative to RIGHT ANTI JOIN using LEFT JOIN*

```
SELECT *
FROM orders AS o
LEFT JOIN customers AS c
    ON c.id = o.customer_id
WHERE c.id IS NULL
```

→ *Alternative INNER JOIN using LEFT JOIN + WHERE*

```
SELECT *
FROM customers AS c
LEFT JOIN orders AS o
    ON c.id = o.customer_id
WHERE o.customer_id IS NOT NULL
```

→ *FULL ANTI JOIN — customers without orders AND orders without customers*

```
SELECT
    c.id,
    c.first_name,
    o.order_id,
    o.customer_id,
    o.sales
FROM customers AS c
FULL JOIN orders AS o
    ON c.id = o.customer_id
WHERE o.customer_id IS NULL OR c.id IS NULL
```

→ *CROSS JOIN — all possible combinations of customers and orders*

```
SELECT *
FROM customers
CROSS JOIN orders
```

3. Multiple Table Joins (4 Tables)

Task: Using SalesDB, retrieve a list of all orders with related customer, product, and employee details. Display: Order ID, Customer name, Product name, Sales amount, Product price, Salesperson name.

```
USE SalesDB
```

```
SELECT
    o.OrderID,
    o.Sales,
    c.FirstName AS CustomerFirstName,
    c.LastName AS CustomerLastName,
    p.Product AS ProductName,
    p.Price,
```

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```
e.FirstName AS EmployeeFirstName,  
e.LastName AS EmployeeLastName  
FROM Sales.Orders AS o  
LEFT JOIN Sales.Customers AS c  
    ON o.CustomerID = c.CustomerID  
LEFT JOIN Sales.Products AS p  
    ON o.ProductID = p.ProductID  
LEFT JOIN Sales.Employees AS e  
    ON o.SalesPersonID = e.EmployeeID
```

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